



STOCK PRECISION PRO

CAPSTONE PROPOSAL PRESENTATION



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OVERVIEW

A Stock Recommender System is currently being developed by our team which aims to provide a seamless and passive experience for users, leveraging industry-specific data, particular and popular ratios sourced from the Yahoo Finance library, stock symbol api, financial models and machine learning techniques to deliver reliable and continuously updated stock recommendations.



NEED ANALYSIS

- **Survey-Driven Demand:**

Insight: Pre-project survey highlights significant demand for stock recommender tool.

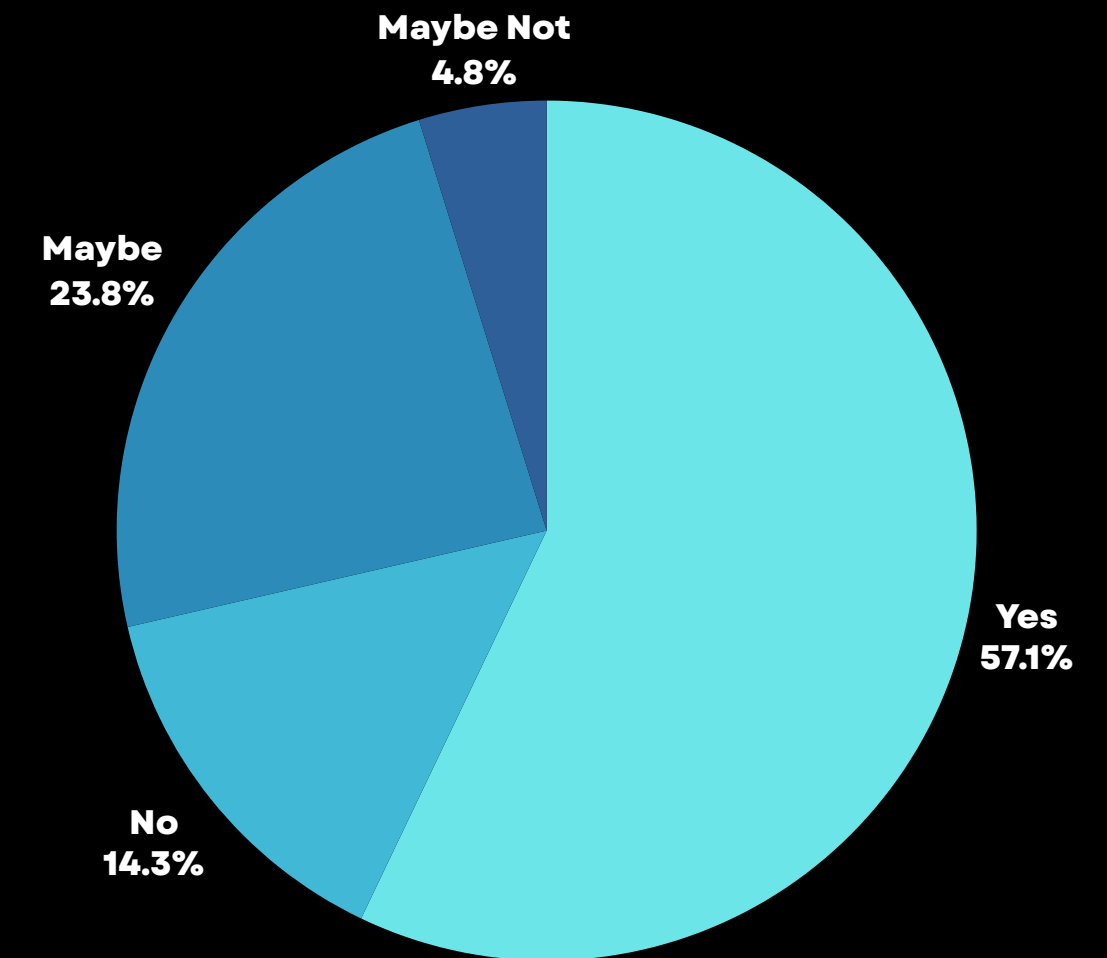
Findings: Many respondents express need for a stock recommendation system regardless of their stock knowledge.

Significance: Aligning project with actual user needs increases adoption and satisfaction.

- **Industry-Centric Sorting:**

Challenge: Generic recommendations lack relevance, tedious for users to filter.

Solution: System sorts recommendations by industry, suggesting specific stocks, simplifying decision-making.



LITERATURE SURVEY

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- Data Sources:

Utilization of Yahoo Finance and alternative web-based APIs for accessing structured historical stock data.

Features evaluated encompass daily prices, trading volumes, volatility, valuation ratios, earnings surprises, and analyst revisions.

- Prediction Models:

Commonly used models such as ARIMA, GARCH, k-nearest neighbours (kNN), and convolutional neural networks (CNN) for stock return data modelling.

- Feature Engineering:

Transformation of raw data into meaningful inputs, including technical indicators and fundamental accounting ratios.

Integration of alternative data like macroeconomic statistics and sentiment analysis from qualitative sources.

- Sector-Focused Recommendations:

Aggregation of stocks into sector buckets for domain-specific benchmarking and fine-tuned recommendation algorithms.

OBJECTIVES

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Algorithm Development: Develop and refine algorithms for sorting stocks based on industry-specific criteria, ensuring the system's capability to provide targeted recommendations in a dynamic market environment.

Real-time Data Integration: Establish a robust system for daily data updates from various sources, including Yahoo Finance library, stock symbol API, and other databases. Ensure seamless integration to keep the sorted stock lists current and reflective of changing market dynamics.

Parameter Optimization: Focus on optimizing the selection and utilization of financial ratios and indicators for dynamic stock evaluation. This objective aims to enhance the effectiveness of the parameters used in assessing the health and performance of companies within their respective industries.

User Experience Enhancement: Evaluate and enhance the user-friendly interface of the website, making it accessible for both seasoned investors and beginners. This objective ensures an intuitive and engaging interaction with the stock recommendation system, fostering a positive user experience.

PROBLEMS

Problem 1

The sheer volume of available financial data can overwhelm users, making it difficult to identify relevant information and make decisions.

Problem 2

Users may be hesitant to invest in the stock market due to fear of losing money or uncertainty about market conditions.

Problem 3

Users may struggle to interpret and apply technical indicators and financial metrics in their investment decisions.

Problem 4

Users may not have the time or expertise to conduct thorough research and analysis of individual stocks.



SOLUTIONS



Solution 1

The solution involves creating a user profile system for investment goals and risk tolerance. Algorithms generate stock recommendations based on these profiles. Default settings and clear explanations simplify the process.

Solution 2

Offer risk assessment tools and portfolio optimization features that help users understand their risk tolerance and diversify their investments accordingly.

Solution 3

Offer customizable stock analysis tools that allow users to filter and compare stocks based on a wide range of criteria, such as valuation multiples, growth rates, and profitability ratios.

Solution 4

Offer automated stock screening and recommendation features based on user-defined criteria, such as investment goals, risk tolerance, and sector preferences.

SERVICES

Service 1

Offer personalized stock recommendations based on user preferences, investment objectives, and market trends



Service 2

Enable users to track the performance of their investments and benchmark against relevant market indices.



Assumptions

- Discount Rate in DCF Model
- Mean Reversion in MACD
- Rate of debt on companies
- Unbiased Sentiments

Constraints

- Data Dependancy
- Delayed data through APIs
- Market volatility
- Risk of model overfitting

METHODOLOGY

Algorithm Development

Gather historical and real-time stock data from multiple sources.
Define industry-specific criteria for sorting stocks.
Develop an adaptable algorithm to sort stocks based on these criteria.



Real-time Data Integration

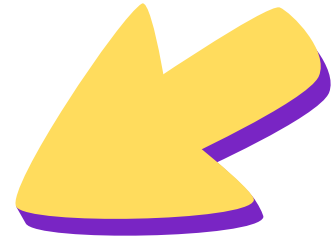
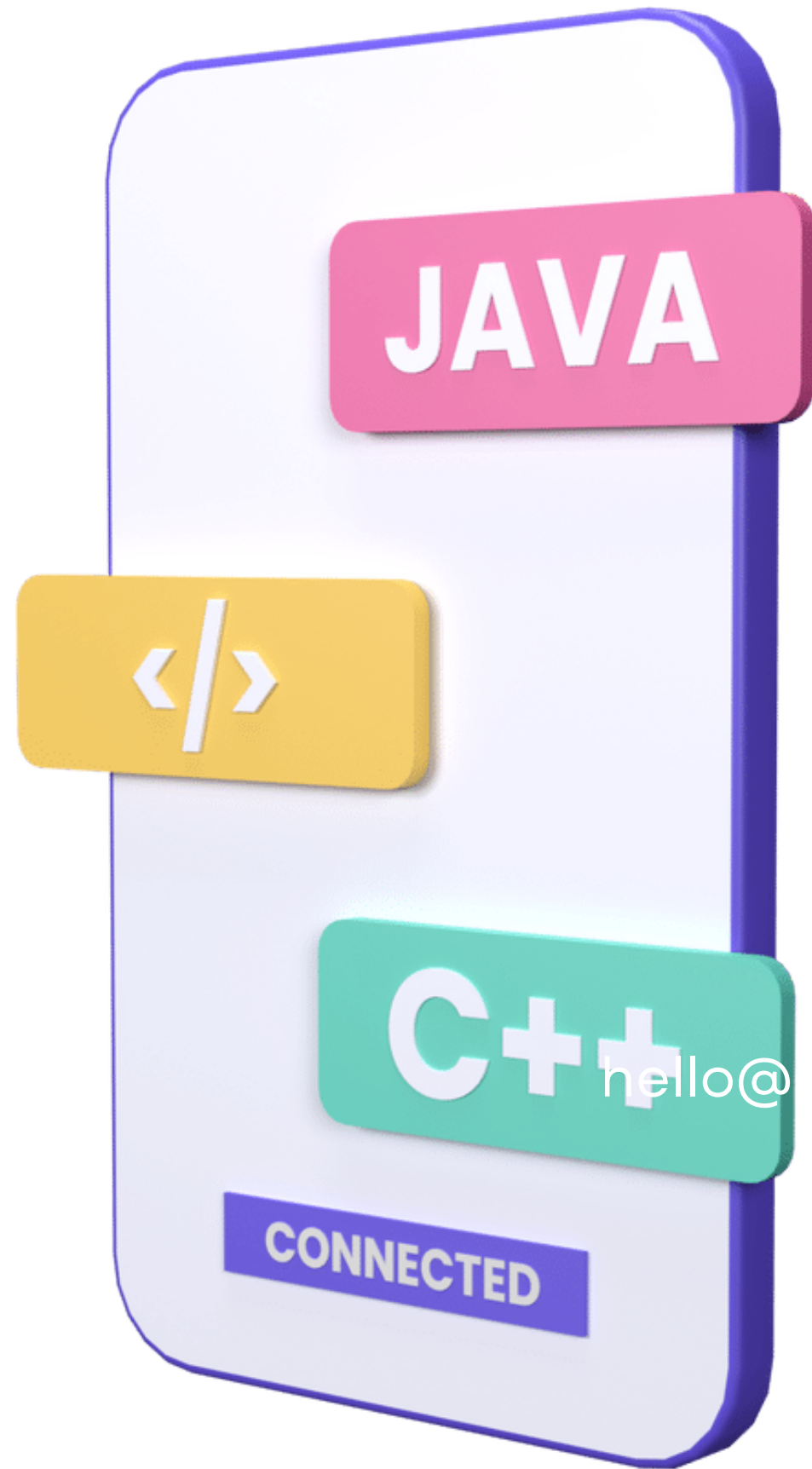
Establish connections to integrate daily data updates seamlessly.
Implement automated processes for data fetching, processing, and updating.
Continuously optimize parameters for evaluating stock performance.



User Experience Enhancement:

Evaluate the user interface for accessibility and user-friendliness.
Incorporate user feedback to improve the interface.
Conduct thorough testing to ensure easy navigation and interaction.





PROJECT REQUIREMENTS

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- PYTHON 3.X
- ELASTICSEARCH 5.X
- KIBANA 5.X
- ELASTICSEARCH PYTHON MODULE
- NLTK PYTHON MODULE
- REQUESTS PYTHON MODULE
- TWEETPY PYTHON MODULE
- BEAUTIFULSOUP4 PYTHON MODULE
- TEXTBLOB PYTHON MODULE
- VADERSENTIMENT PYTHON MODULE
- NEWSPAPER3K PYTHON MODULE

PROJECT OUTCOMES



Stock market analysis by machine learning models, industry centric stock recommendations and decision making capabilities.

Real time updates of the system as a dynamic tool adapting to changing market trends and effective investment choices

Project Head

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Finance Analyst

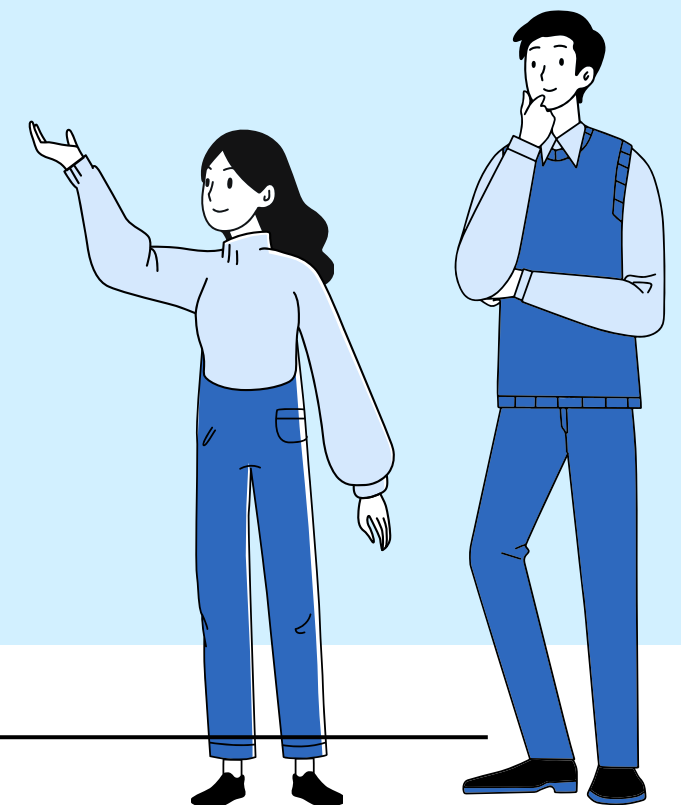
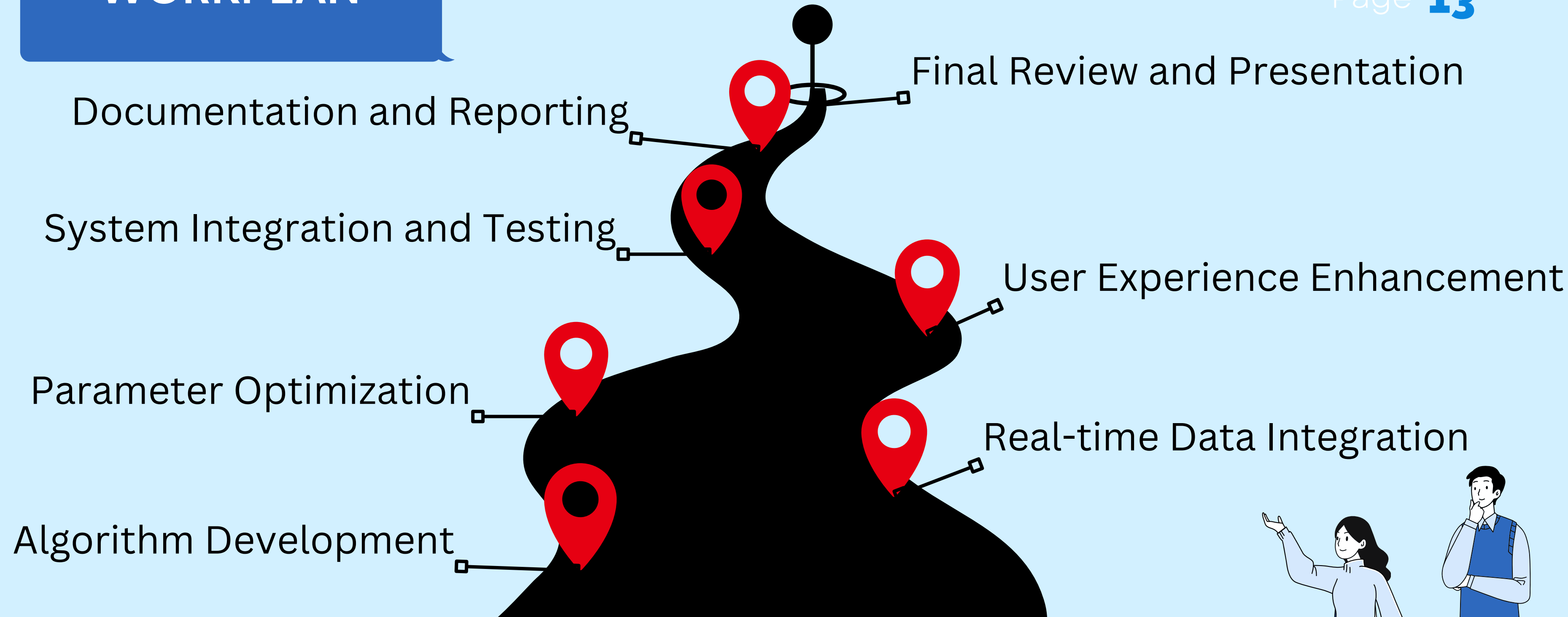
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**THANK
YOU!**
